



Sundaresan Ravi

📍 **Home:** No: 20 Dhanalakshmi Nagar, Walajabad, Kancheepuram, 631605, Kancheepuram, India

✉ **Email address:** rs2627@srmist.edu.in

🌐 **Website:** <https://sundaresanravi-aichem.github.io/>

🌐 **LinkedIn:** www.linkedin.com/in/sundaresanravi

ABOUT MYSELF

Chemistry graduate (Honors) with hands-on experience in quantum chemical calculations, molecular machine learning, and computational spectroscopy. Developed machine learning models to predict spectroscopic properties of transition-metal complexes using molecular descriptors as computationally efficient alternatives to explicit quantum chemical calculations. Experienced in dataset curation, descriptor engineering, and predictive modeling for molecular property prediction. Seeking a PhD position at the intersection of quantum chemistry, spectroscopy, and data-driven molecular and materials discovery.

EDUCATION & TRAINING

[12/08/2022 - 13/04/2026]

Bachelors of Sciences (B.S.) (Honors)

SRM Institute of Science and Technology - Kattankulathur

City: Kattankulathur | **Country:** India | **Field(s) of study:** Chemistry | **Level in EQF:** 6

Cumulative GPA: 8.8 / 10

Coursework: Quantum Chemistry and Group Theory · Classical and Statistical Thermodynamics · Molecular Spectroscopy and Photochemistry · Electrochemistry and Chemical Dynamics · Solid State Chemistry · Introduction to Computational Chemistry (Elective) · Advanced Physical Chemistry (Honors Track) · Mathematics (Calculus, Statistics).

WORK EXPERIENCE

[21/04/2025 - 21/07/2025]

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

City: Kattankulathur | **Country:** India

Research intern

Enhancing Supercapacitive Performance of WS_2 via Transition-Metal Intercalation

Supervisor: Dr. Tumpa Sadhukhan, Research Assistant Professor

Description of activities and responsibilities:

Performed periodic DFT calculations using VASP to study the structural stability, electronic structure, and energetics of transition-metal-intercalated WS_2 systems, developed Python-based workflows to process density-of-states (DOS) data and compute quantum capacitance, and analyzed structure-property relationships relevant to supercapacitor performance.

LANGUAGE SKILLS

Mother tongue(s): Tamil

Other language(s):

English

LISTENING: C2 **READING:** C1 **WRITING:** B2

SPOKEN PRODUCTION: B2 **SPOKEN INTERACTION:** B2

SKILLS

Microsoft Office | LaTeX | Git | Density Functional Theory (DFT) | Computational Materials Science | VASP | Python (computer programming) | Origin | Cheminformatics | machine learning | RDKit

PROJECTS

[02/01/2025 - 09/06/2025]

Machine learning analysis for binding-energy prediction on functionalized borophene.

- Curated and preprocessed a dataset of DFT-computed binding energies for functionalized borophene surfaces, integrating structural and electronic descriptors for machine-learning analysis.
- Developed, optimized, and benchmarked **regression models** for binding-energy prediction, enabling identification of key structure–property relationships.

[02/12/2025 - Current]

Machine learning for predicting spectroscopic properties of transition-metal complexes.

- **Designed** dative-bonded SMILES representations of transition-metal complexes for machine-learning input.
- **Built & curated** a database of transition-metal complexes and performed quantum chemical calculations(*Gaussian*) to extract electronic and spectroscopic properties.
- **Prepared** datasets and computational workflows for machine-learning models to predict spectroscopic behavior relevant to photodynamic therapy.

CONFERENCES & SEMINARS

[25/08/2023 - 26/08/2023]

ASTON's Workshop on Mass Spectrometry SRM Institute of Science and Technology

[21/02/2024 - 22/02/2024]

Computational Methods for Solving Chemical Problems National Institute of Technology, Trichy

[07/06/2024 - 09/06/2024]

AI and Machine Learning in Material Science Indian Institute of Technology, Madras

[01/09/2024 - 07/09/2024]

Hands-On Workshop on Molecular Docking & MD Simulation: From Theory and Practical Online

[01/09/2024 - 07/09/2024]

Hands-On Workshop on Density Functional Theory (DFT) Calculations Online

[18/01/2025 - 19/01/2025]

Computational Chemistry Techniques – NPTEL+ Online

[20/01/2025 - 24/01/2025]

Materials Informatics: Fingerprints, GNNs and Transformer Models for Materials Design Indian Institute of Technology, Madras

CERTIFICATIONS

[Udemy, 10/12/2025]

QC101 Quantum Computing & Intro to Quantum Machine Learning

Introductory course covering quantum computing fundamentals and basic concepts of quantum machine learning.

Mode of Learning: Online

[British Council / IDP Education / Cambridge Assessment English, 23/11/2025]

IELTS Academic

Listening 8.0, Reading 7.0, Writing 6.5, Speaking 6.5
(TEST REPORT NUMBER: 25IA531393RAVS008A)